

64+1

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Midterm Examination  
CHE 230  
Molecular Systems in Chemical Engineering

UIC Chemical Engineering Department

03/02/2026

Time: 70 minutes

Honor Code: As a commitment to the personal, academic, and professional standards upheld by the Honors College, as a student, I hereby pledge to respect, preserve, and promote the Honor Code of Conduct during my entire tenure at the Honors College. I agree to uphold high ethical principles in all personal, academic, and professional endeavors related to the Honors College and beyond, and to always be mindful of being a representative of the Honors College and the UIC community

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Date 03/02/26

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Section 1: Fundamental to Molecular Systems (30 points)

1. (5 points) What is the difference between a physical and a chemical<sup>bond</sup>? Give one example of each. A chemical bond describes a change in chemical makeup, a physical bond describes 2 objects bonded together. A chemical bond would be  $\text{Na}^{\oplus} + \text{Cl}^{\ominus} \rightarrow \text{NaCl}$ , a physical bond would be like welding 2 pieces of metal together.

X |

2. (5 points) Graphite and diamond are chemically identical yet have different physical properties. Please briefly explain why.

It all has to do with the way the atoms are arranged. In diamonds, the C atoms are uniform and in graphite they are not



Diamond



Graphite

3

3. (5 points) Briefly cite the main differences between ionic, covalent, and metallic bonding.

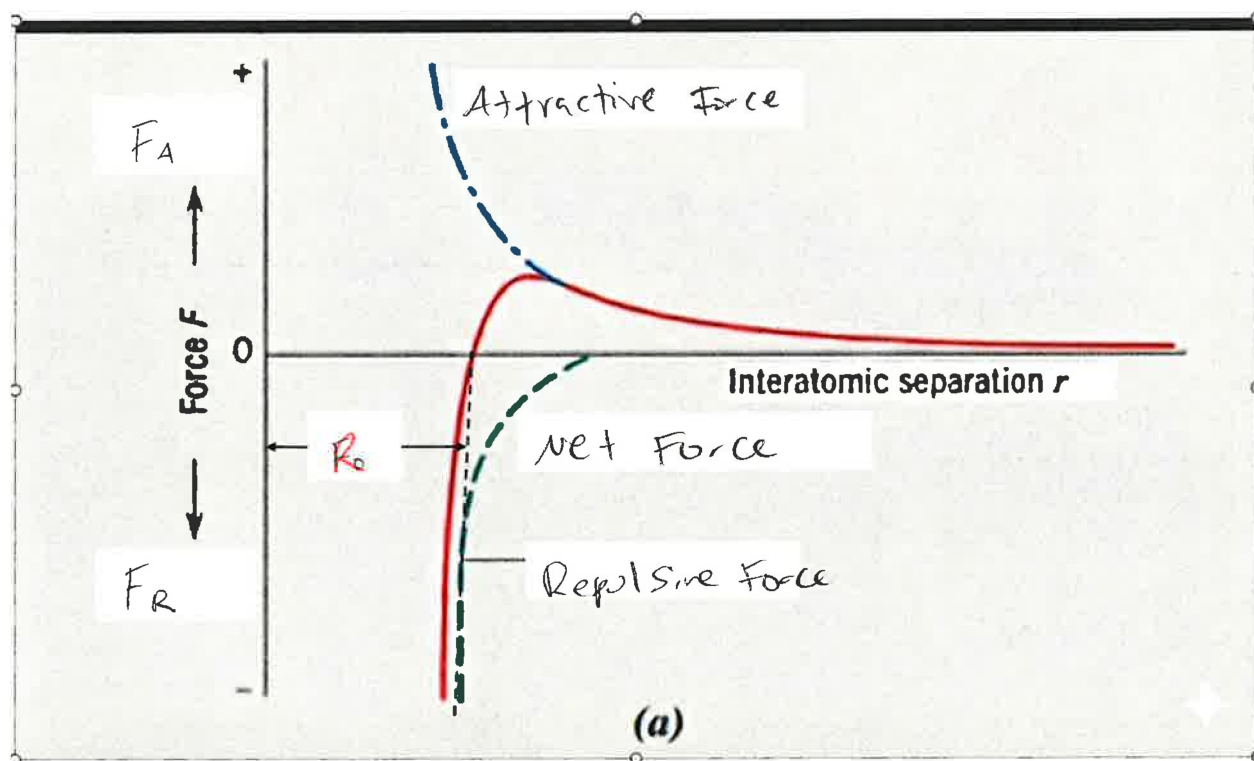
Ionic :  $\begin{matrix} \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus \end{matrix}$  all positive atoms

Covalent :  $\text{O} \text{---} \text{O}$   $e^-$  overlap

Metallic :  $\begin{matrix} \oplus \ominus \oplus \\ \ominus \oplus \ominus \\ \oplus \ominus \oplus \end{matrix}$  , positive + negative alternating

4. (6 points) The graph provided illustrates the forces acting between two atoms as a function of their interatomic separation,  $r$ .

Fill in the blank boxes on the diagram with the correct terms.



5. (9 points) The atomic radii of  $\text{Mg}^{2+}$  and  $\text{O}^{2-}$  ions are 0.072 nm and 0.140 nm, respectively.

(a) Calculate the force of attraction between these two ions at their equilibrium interionic separation (i.e., when the ions just touch one another) 6

(b) What is the force of repulsion at this same separation distance? 3

4+1

Given Constants:

- Magnitude of electrical charge for an electron (e):  $1.602 \times 10^{-19} \text{ C}$
- Permittivity of a vacuum  $\epsilon_0$ :  $8.854 \times 10^{-12} \text{ F/m}$  (or  $\text{C}^2/(\text{N}\cdot\text{m}^2)$ )

a)

$$\text{Mg}^{2+} : r_1 = 0.072 \text{ nm}$$

$$\text{O}^{2-} : r_2 = 0.140 \text{ nm}$$

$$r = r_1 + r_2 = 0.212 \text{ nm}$$

$$F_A = \frac{1}{4\pi\epsilon_0 r^2} (|z_1|e)(|z_2|e) = \frac{1}{4\pi\epsilon_0 (0.212 \text{ nm})} (2e)(2e)$$

given const.

$$|z_1| = |+2| = 2$$

$$|z_2| = |-1| = 2$$

$$F_A = 2.17 \times 10^{-27} \text{ N}$$

$$b) F_A + F_R = 0, F_R = -F_A, F_R = -2.17 \times 10^{-27} \text{ N}$$

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## Section 2: Crystal Solid (25 points)

$$\rho = \frac{\text{g}}{\text{cm}^3} \rightarrow \frac{\text{g}}{\text{mol}} \cdot \frac{\text{mol}}{\text{cm}^3}$$

1. (5 points) Calculate the radius of an iridium atom, given that Ir has an FCC crystal structure, a density of  $22.4 \text{ g/cm}^3$ , and an atomic weight of  $192.2 \text{ g/mol}$  ( $N_A = 6.022 \times 10^{23} \text{ atoms/mol}$ ).

$$\rho = \frac{N_A}{V_c \times N_A}$$

4 solve  $V_c$

$$V = \frac{4}{3} \pi r^3$$

$$22.4 \frac{\text{g}}{\text{cm}^3} = \frac{192.2 \text{ g/mol}}{V_c \times 6.022 \times 10^{23}}$$

$$V_c \times 6.022 \times 10^{23} = \frac{192.2}{22.4}$$

$$V_c = 1.42 \times 10^{-23} = \frac{4}{3} \pi r^3$$

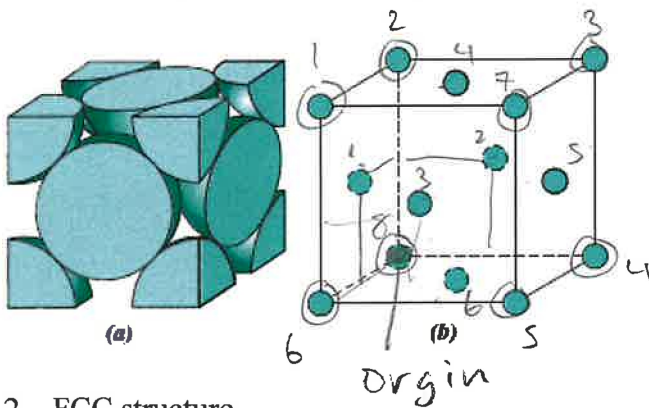
edge length

$$r = 1.84 \times 10^{-12} \text{ cm}$$

we need units  $\text{cm}^3$   $\frac{\text{g}}{\text{mol}} \cdot \frac{\text{mol}}{\text{atom}} = \left(\frac{\text{g}}{\text{atom}}\right) \rightarrow$  this is correct

2. (10 points) List the point coordinates for all atoms that are associated with the FCC unit cell (Figure 2)

points (x, y, z)



$$8 + 6 = 14 \text{ total}$$

✓ 10

Figure 2 – FCC structure

Corners

|   |           |   |           |
|---|-----------|---|-----------|
| 1 | (1, 0, 1) | 5 | (1, 1, 0) |
| 2 | (0, 0, 1) | 6 | (1, 0, 0) |
| 3 | (0, 1, 1) | 7 | (1, 1, 1) |
| 4 | (0, 1, 0) | 8 | (0, 0, 0) |

face

|   |               |   |               |
|---|---------------|---|---------------|
| 1 | (1/2, 0, 1/2) | 4 | (1/2, 1/2, 1) |
| 2 | (0, 1/2, 1/2) | 5 | (1/2, 1, 1/2) |
| 3 | (1, 1/2, 1/2) | 6 | (1/2, 1/2, 0) |



3. (5 points) What is the unit cell number, coordination number, Edge Length and APF of a Body Centered Cubic (Show calculation of APF).

BCC

Unit cell #:  $N = N_c + \frac{N_F}{2} + \frac{N_C}{8} = 1 + 0 + 1 = 2$  ✓

CO #: 8 ✓

Edge length:  $\frac{4}{\sqrt{3}} r$  ✗

APF =  $\frac{V_{atoms}}{V_{cube}} = \frac{\frac{4}{3}\pi r^3 \times 2}{\left(\frac{4}{\sqrt{3}} r\right)^3} = \frac{2 \times \frac{4}{3}\pi}{\frac{4}{\sqrt{3}}} = 0.68$  ✓

4. (5 points) Match each definition or description below with the most appropriate term from the word bank. Each term may be used once, more than once, or not at all.

**Word Bank**

|                              |                                |                       |                       |
|------------------------------|--------------------------------|-----------------------|-----------------------|
| <b>Grain Boundary</b>        | <b>Substitutional impurity</b> | <b>Single Crystal</b> | <b>Vacancy Defect</b> |
| <b>Interstitial Impurity</b> | <b>Coordination Number</b>     | <b>Edge Length</b>    | <b>Unit Cell</b>      |

Coordination # The number of nearest neighbor or touching atom. ✓

Grain Boundary A two-dimensional planar defect that separates regions of identical crystal structure but differing crystallographic orientations. ✓

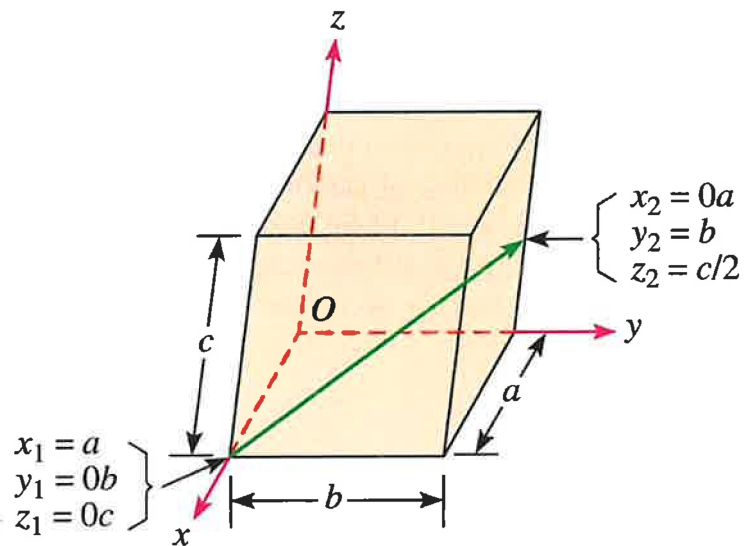
Substitutional impurity Foreign atom replaces host atom. ✓

Edge Length The physical dimension or length of one side of a unit cell. ✓

Single crystal A crystalline solid in which the perfectly ordered, repeating arrangement of atoms is continuous and unbroken throughout the entire sample. ✓

**Extra Credit**

**(5 points) Determine the indices for the direction shown in the accompanying figure.**



$[0 \ 1 \ 0.5]$

Direction:  $[u \ v \ w]$

$[1 \ 0 \ 0]$   $\times$

$[\bar{2}, 2, 1]$

### Section 3. Material Properties (45 points)

26

#### Diffusion

1. (10 points) For some applications, it is necessary to harden the surface of a steel (or iron-carbon alloy) above that of its interior. One way this may be accomplished is by increasing the surface concentration of carbon in a process termed carburizing; the steel piece is exposed, at an elevated temperature, to an atmosphere rich in a hydrocarbon gas, such as methane ( $\text{CH}_4$ ). Consider one such alloy that initially has a uniform carbon concentration of 0.25 wt% and is to be treated at 950°C (1750°F). If the concentration of carbon at the surface is suddenly brought to and maintained at 1.20 wt%, how long will it take to achieve a carbon content of 0.80 wt% at a position 0.5 mm below the surface? The diffusion coefficient for carbon in iron at this temperature is  $1.6 \times 10^{-11} \text{ m}^2/\text{s}$ ; assume that the steel piece is semi-infinite.

Error Function is on the last page

$$\frac{C_i - C_x}{C_x - C_s} = \frac{1}{\sqrt{\pi}} \operatorname{erf}\left(\frac{\sqrt{D \times T}}{h}\right)$$

$$\frac{C(x) - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\frac{0.25 - 0.8}{0.8 - 1.20} = \frac{1}{\sqrt{\pi}} \operatorname{erf}\left(\frac{\sqrt{1.6 \times 10^{-11} \times 1750}}{0.0005}\right)$$

$\rightarrow 0.112$

2

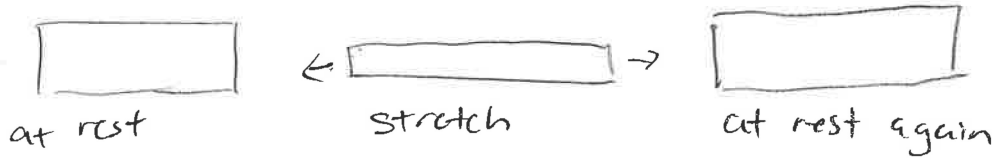
$$1.375 = \operatorname{erf}(0.1236 \dots)$$

$$\underline{t = 11.2 \text{ hrs}}$$

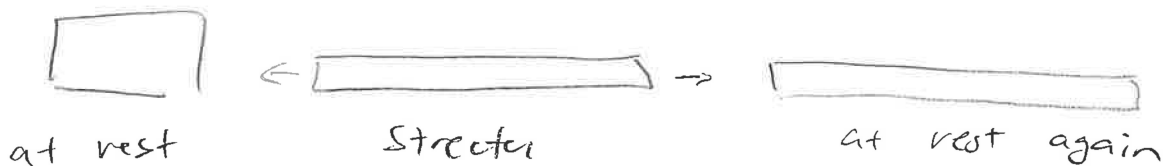
X

2. (4 points) **Distinguish** between elastic and plastic deformations, both by definition, and in terms of behavior on a stress-strain plot. **Draw** a sketch to illustrate your answer.

elastic: you stretch it + it goes back to original state



plastic: you stretch it + it does NOT go back to original state



3. (3 points) **Highlight** the differences between true stress and engineering stress.

True stress:  $\sigma_T = \frac{F}{A}$  Eng Stress:  $\sigma_E = \frac{F}{A_0}$

True stress accounts for any elongation in the material, whereas engineering stress does not.

True stress uses instantaneous area.

4. (8 points) Of the metals listed in the Table below,

- (a) Which will experience the greatest percent reduction in area? Why?
- (b) Which is the strongest? Why?
- (c) Which is the stiffest? Why?
- (d) Which is the hardest? Why?

| Material | Yield Strength (MPa)      | Tensile Strength (MPa) | Strain at Fracture | Fracture Strength (MPa) | Elastic Modulus (GPa) |
|----------|---------------------------|------------------------|--------------------|-------------------------|-----------------------|
| A        | 310                       | 340                    | 0.23               | 265                     | 210                   |
| B        | 100                       | 120                    | 0.40               | 105                     | 150                   |
| C        | 415                       | 550                    | 0.15               | 500                     | 310                   |
| D        | 700                       | 850                    | 0.14               | 720                     | 210                   |
| E        | Fractures before yielding |                        |                    | 650                     | 350                   |

a) Material B, because it has the highest Strain at fracture

b) Material D, because it has the highest tensile strength

c) Material E, because it has the highest Elastic Modulus

d) Material D, because it has the highest yield strength

✓  
8

5. (10 points) Consider a cylindrical titanium wire 3.0 mm (0.12 in.) in diameter and  $2.5 \times 10^4$  mm (1000 in.) long. When a load of 500 N (112 lbf) is applied, the wire experiences an elongation of 16.5 mm. Assuming that the deformation is totally elastic, calculate the Young's Modulus of the titanium in GPa.

$$d = 3 \text{ mm} \quad l = 2.5 \times 10^4 \text{ mm} \quad F = 500 \text{ N}$$

$$\text{elongation} = 16.5 \text{ mm} \Rightarrow \Delta L$$

$$\Delta L = \frac{F \times L}{A \times E} = \frac{500 \text{ N} \times 2.5 \times 10^4 \text{ mm}}{2.356 \times 10^5 \times E} = 16.5 \text{ mm}$$

Area  $\pi r^2$

$$A = 2\pi RL$$

$$\frac{2.356 \times 10^5 \times E}{500 \times 2.5 \times 10^4} = \frac{1}{16.5}$$

2.

$$E = 3.0876 \text{ Pa} \quad \times$$

- b. A single crystal of some metal has a  $\tau_{\text{crss}}$  of 30.7 MPa and is exposed to a tensile stress of 61 MPa.

(2 points) (i) Will yielding occur when  $\phi = 56^\circ$  and  $\lambda = 32^\circ$ .

(2 points) (ii) If not, what stress is necessary?

$$(i) \quad \tau_R = \sigma \cos \phi \cos \lambda = (61) \cos(56) \cos(32)$$

$$\tau_R = 28.93 \text{ MPa} \quad 28.93 < 30.7 \quad \checkmark \quad 2$$

yielding will not occur

$$(ii) \quad 30.7 = \sigma \cos(56) \cos(32)$$

$$\sigma = 64.74 \text{ MPa} \quad \checkmark \quad 2$$

6. (6 points) A relatively large plate of a glass is subjected to a tensile stress of 40 MPa. If the specific surface energy and modulus of elasticity for this glass are  $0.3 \text{ J/m}^2$  and  $69 \text{ GPa}$  respectively, determine the maximum length of a surface flaw that is possible without fracture.

$$\text{Surface energy} = 0.3 \text{ J/m}^2$$

$$\text{modulus of el.} = 69 \text{ GPa}$$

$$\sigma = 40 \text{ MPa}$$

$$\sigma_c = \sqrt{\frac{2E\gamma_s}{\pi a}} \quad \rightarrow \quad 40 = \sqrt{\frac{2(69)(0.3)}{\pi a}}$$

$\nearrow$  solve

$$(40)^2 = \left( \sqrt{\frac{41.4}{\pi a}} \right)^2$$

$$1600 = \frac{41.4}{\pi a}$$

$$\frac{\pi a}{\pi} = \frac{41.4}{1600} = \frac{0.025875}{\pi}$$

$$\underline{a = 0.008236 \text{ m}}$$

✓  
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## Appendix G: Tables of Error Function

| X   | ERF           | X   | ERF           | X    | ERF           | X    | ERF           |
|-----|---------------|-----|---------------|------|---------------|------|---------------|
| .01 | .011283415556 | .46 | .484655390002 | .91  | .801882825766 | 1.36 | .945561436561 |
| .02 | .022564574692 | .47 | .493745050886 | .92  | .806767721548 | 1.37 | .947312398035 |
| .03 | .033841222342 | .48 | .502749670695 | .93  | .811563558585 | 1.38 | .949016035256 |
| .04 | .045111106145 | .49 | .511668261189 | .94  | .816271018976 | 1.39 | .950673295805 |
| .05 | .056371977797 | .50 | .520499877813 | .95  | .820890807273 | 1.40 | .952285119763 |
| .06 | .067621594393 | .51 | .529243619841 | .96  | .825423649644 | 1.41 | .953852439360 |
| .07 | .078857719771 | .52 | .537898630479 | .97  | .829870293036 | 1.42 | .955376178641 |
| .08 | .090078125841 | .53 | .546464096935 | .98  | .834231504340 | 1.43 | .956857253145 |
| .09 | .101280593915 | .54 | .554939250456 | .99  | .838508069555 | 1.44 | .958296569601 |
| .10 | .112462916018 | .55 | .563323366325 | 1.00 | .842700792950 | 1.45 | .959693023637 |
| .11 | .123622896199 | .56 | .571615763824 | 1.01 | .846810496228 | 1.46 | .961053509513 |
| .12 | .134758351820 | .57 | .579815806164 | 1.02 | .850838017701 | 1.47 | .962372899854 |
| .13 | .145867114836 | .58 | .587922900382 | 1.03 | .854784211454 | 1.48 | .963654065414 |
| .14 | .156947033063 | .59 | .595936497198 | 1.04 | .858649946527 | 1.49 | .964897864843 |
| .15 | .167995971427 | .60 | .603856090848 | 1.05 | .862436106090 | 1.50 | .966105146475 |
| .16 | .17901181319  | .61 | .611681218876 | 1.06 | .866143586635 | 1.51 | .967276748129 |
| .17 | .189992461202 | .62 | .619411461899 | 1.07 | .869773297164 | 1.52 | .968413496920 |
| .18 | .200935839019 | .63 | .627046443338 | 1.08 | .873326158388 | 1.53 | .969516209093 |
| .19 | .211839892158 | .64 | .634585829122 | 1.09 | .876803101938 | 1.54 | .970585689861 |
| .20 | .222702589210 | .65 | .642029327356 | 1.10 | .880205069574 | 1.55 | .971622733262 |
| .21 | .233521922982 | .66 | .649376687963 | 1.11 | .883533012415 | 1.56 | .972628122027 |
| .22 | .244295911599 | .67 | .656627702300 | 1.12 | .886787890165 | 1.57 | .973602627462 |
| .23 | .255022599592 | .68 | .663782202741 | 1.13 | .889970670363 | 1.58 | .974547009343 |
| .24 | .265700058954 | .69 | .670840062235 | 1.14 | .893082327630 | 1.59 | .975462015822 |
| .25 | .276326390168 | .70 | .677801193837 | 1.15 | .896123842937 | 1.60 | .976348383345 |
| .26 | .286899723216 | .71 | .684665550217 | 1.16 | .899096202880 | 1.61 | .977206836583 |
| .27 | .297418218547 | .72 | .691433123139 | 1.17 | .902000398966 | 1.62 | .978038088373 |
| .28 | .307880068029 | .73 | .698103942917 | 1.18 | .904837426915 | 1.63 | .978842839674 |
| .29 | .318283495861 | .74 | .704678077855 | 1.19 | .907608285972 | 1.64 | .979621779524 |
| .30 | .328626759459 | .75 | .711155633654 | 1.20 | .910313978230 | 1.65 | .980375585023 |
| .31 | .338908150311 | .76 | .717536752806 | 1.21 | .912955507973 | 1.66 | .981104921311 |
| .32 | .349125994796 | .77 | .723821613965 | 1.22 | .915533881027 | 1.67 | .981810441565 |
| .33 | .359278654974 | .78 | .730010431299 | 1.23 | .918050104127 | 1.68 | .982492787002 |
| .34 | .369364529345 | .79 | .736103453821 | 1.24 | .920505184299 | 1.69 | .983152586895 |
| .35 | .379382053562 | .80 | .742100964708 | 1.25 | .922900128256 | 1.70 | .983790458591 |
| .36 | .389329701129 | .81 | .748003280598 | 1.26 | .925235941810 | 1.71 | .984407007545 |
| .37 | .399205984043 | .82 | .753810750875 | 1.27 | .927513629295 | 1.72 | .985002827359 |
| .38 | .409009453420 | .83 | .759523756938 | 1.28 | .929734193014 | 1.73 | .985578499828 |
| .39 | .418738700070 | .84 | .765142711455 | 1.29 | .931898632689 | 1.74 | .986134594997 |
| .40 | .428392355047 | .85 | .770668057608 | 1.30 | .934007944941 | 1.75 | .986671671219 |
| .41 | .437969090155 | .86 | .776100268324 | 1.31 | .936063122773 | 1.76 | .987190275231 |
| .42 | .447467618426 | .87 | .781439845491 | 1.32 | .938065155079 | 1.77 | .987690942224 |
| .43 | .456886694550 | .88 | .786687319174 | 1.33 | .940015026158 | 1.78 | .988174195930 |
| .44 | .466225115278 | .89 | .791843246814 | 1.34 | .941913715258 | 1.79 | .988640548706 |
| .45 | .475481719787 | .90 | .796908212423 | 1.35 | .943762196123 | 1.80 | .989090501636 |

